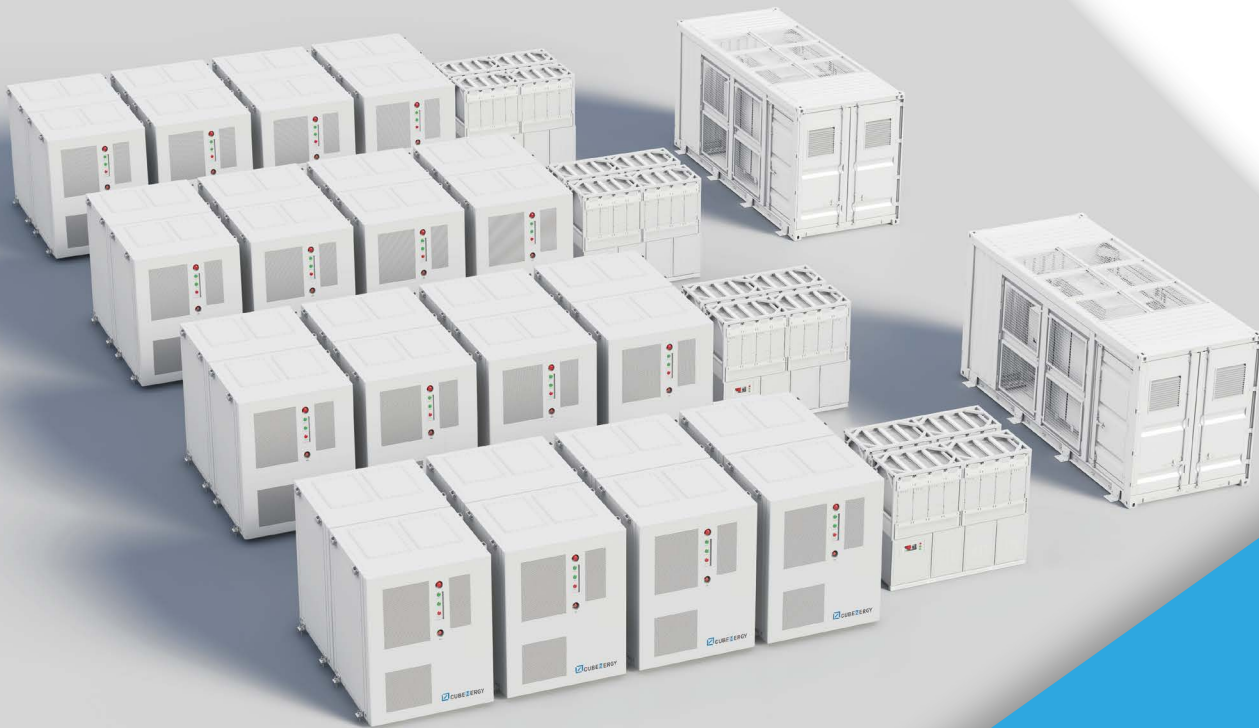




FlexCombo

Flexible, Scalable & Reliable Energy Block

FlexCombo



FlexCombo is a reliable, intelligent and fully-integrated battery energy storage system developed by CUBENERGY for applications in wide-range projects conditions and locations.

Modular design & parallel-connected method of FlexCombo indicate more flexibility and effortless-experience by shortening the delivery period and simplifying the deployment process.



All-in-One & Plug-and-play

Ease of transportation, installation, operation and maintenance;



Flexible Scalability

Simply deploying & connecting standardized blocks to increase energy capacity;



Intelligent Management

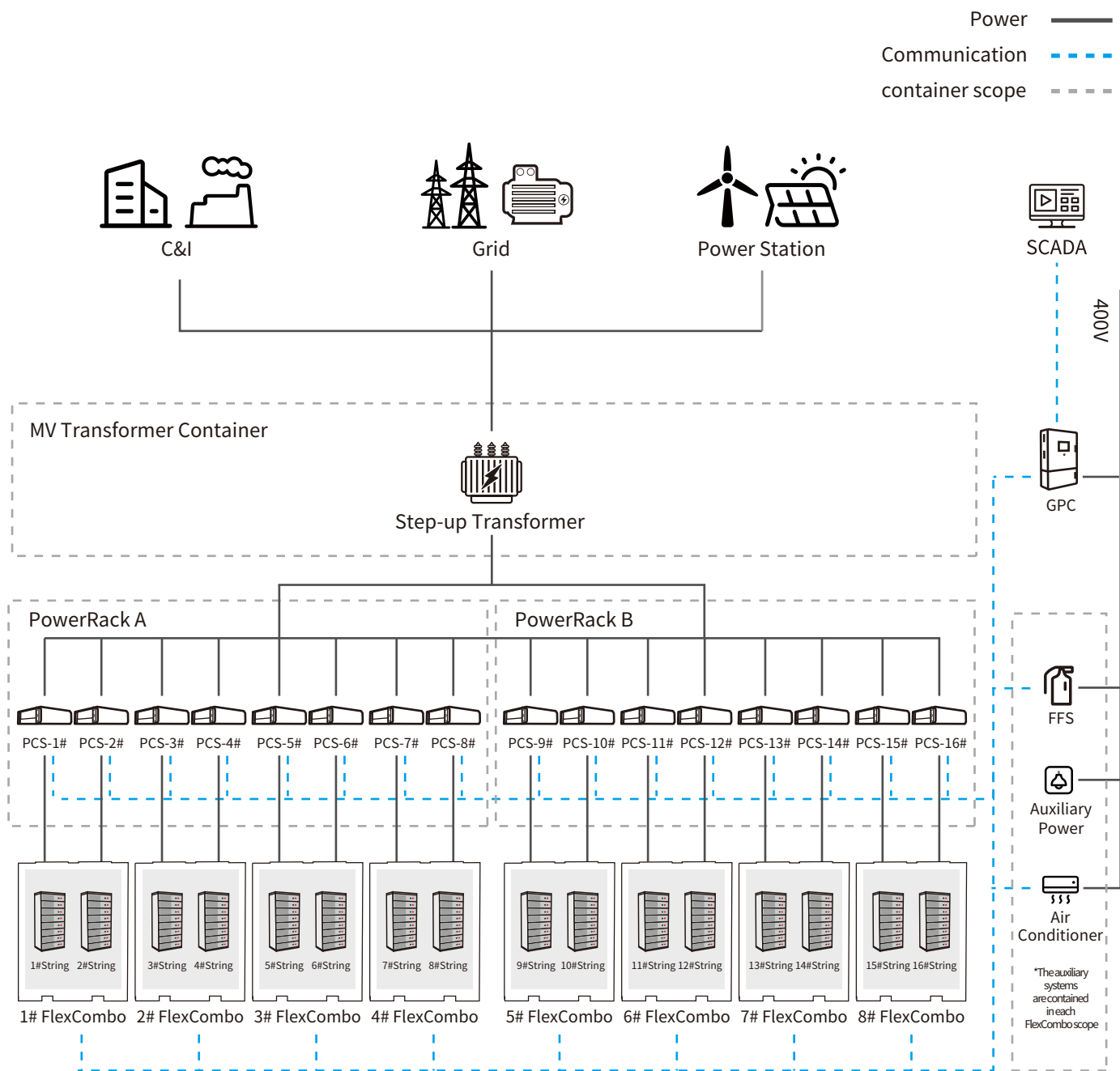
Integrated with leading balancing technology & operation platform, huge plus for safety and long-life-span.



FlexCombo is designed with an easy augmentation feature to adapt evolving project requirements. Its DC/AC one-on-one topology, combined with a modular design, simplifies AC augmentation by enabling the addition of PCS units and new enclosures as needed.

This flexible approach supports incremental expansion of the system. By integrating two standardized FlexCombo units and two additional PCS to at each stage, the system's capacity can be scaled up from 6,687kWh, ensuring at least 10 years of stable projects operation.

System Topology



Product Configuration

Product Model	Battery String Type	FlexCombo Qty	Nominal Capacity	DC Voltage Range	PCS Grid-connected Voltage
FlexCombo - D6500	LS417	8	6,687kWh	1,164.8V~1,476.8V	690V

System Technical Specifications

Item	FlexCombo-D6500
DC Data	
Battery chemistry	Lithium Iron Phosphate (LFP)
Number of D1 (DC block)	8
DC rated energy capacity	6,687kWh
Rated voltage	1,331.2V
Voltage range	1,164.8V~1,476.8V
Auxiliary load voltage	400V, 50/60Hz
Auxiliary peak load	54kW
BMS communication interface	RS485, Ethernet
BMS communication protocol	Modbus RTU, Modbus TCP
AC Data	
Rated AC Power	3,200kW
Maximum AC power	3,520kW
Rated voltage	690V
Grid voltage range	586.5~759V (Optional)
AC rate of current	2,677.6A
Output THDi	<3%
AC PF	-1~+1
AC output	3W+PE
General Data	
Degree of protection	IP54
Operating temperature range	-30~50°C
Relative humidity	0~95% (non-condensing)
Max working altitude	<2,000m/6,561ft
Cooling concept of DC hatch	Liquid Cooling
Coolant type	50% Ethylene glycol solution
Fire fighting system	Aerosol/Water fire fighting
Communication interfaces	RS485, Ethernet
Certificates	UL9540, CE

Key Components



Flexcombo

Item	D1
Battery chemistry	Lithium Iron Phosphate (LFP)
Cell life cycle	8,000 cycles with 70% retention @ 0.5C 25°C
Cell spec	3.2V/314Ah
Each string configuration	1P416S
Number of strings	2
DC rated energy capacity	836kWh
Rated voltage	1,331.2V
Voltage range	1,164.8V~1,476.8V
Auxiliary load voltage	400V, 50/60Hz
Auxiliary peak load	6.7kW
BMS communication interface	RS485, Ethernet
BMS communication protocol	Modbus RTU, Modbus TCP
Dimension (W*D*H)	2,000mm*1,785mm*2,545mm
Weight of the whole system	7.8t

□ Key Components



Power Conversion System

Item	EH-0200-HA-M
Rated AC power	200kW
Max.DC current	224.5A
DC voltage range	1000~1500Vd.c
Max.DC voltage	1500Vd.c
Power factor range	-1~+1
Ingress protection	IP66
Rated AC current	167.4A
Max.AC current	184.1A
Rated AC voltage	690Va.c.(3W+PE)
Output operating voltage range	-15%~+10%(Configurable)
Rated AC frequency	50/60Hz
Operating temperature range	-40~+60°C
Protective class	I
Operating altitude	4000m(Derating above 3000 m)
Cooling method	Smart air cooling
Weight	100kg



PowerRack-EU

Item	
DC Parameters	
Max.DC current	224.5A*8
DC voltage range	1,000-1,500V
Number of DC inputs	8
AC parameters (On-Grid)	
Rated output power	200kW*8
Rated grid voltage	690V, 3W+PE
Output operating voltage range	-15%~+10%(Configurable)
Grid frequency	50Hz/60Hz
Rated AC current	167.4A*8
AC PF	-1~+1
Dimensions (W*D*H)	2,550mm*900mm*2,100mm
Weight	≤2t

- Single-stage three-level modularization;
- Multi-branch input to reduce battery series and parallels connection;
- One-to-one Management of Battery string and PCS.

□ Key Components



Certificates: IEC 62271-202, EN 50588-1, IEC 60076, IEC 62271-200, IEC 61439-1

MV Transformer

Item	CBMV-3300-EU		CBMV-6600-EU	
Input				
Rated Input Voltage	690V			
Phase number	3Φ			
AC Power	3300kVA@40°C		6600kVA@40°C	
LV Main Switches	ACB (1,600A/3P)×2		ACB (1,600A/3P)×4	
Output				
Rated Output Voltage	10~35kV			
Frequency	50Hz/60Hz			
Transformer Tappings	±2 x 2.5%			
Transformer Cooling Type	ONAN		ONAN	
Transformer Oil Type	Mineral Oil (PCB Free)		Mineral Oil (PCB Free)	
Transformer Vector Group	Dy11		Dy11-y11	
Transformer Min. Peak Efficiency Index	Tier 2 In Accordance with EN 50588-1			
HVC Type	SF6 Gas Insulated			
HVC Transformer Protection Unit	MV Vacuum circuit breaker			
Transformer Type	Oil-immersed Transformer		Oil-immersed Transformer	
Auxiliary Transformer type	SGL-100kVA 0.69/0.4kV Dyn11		SGL-100kVA 0.69/0.4kV Dyn11	
Output Voltage of Auxiliary Transformer	400Vac			
General				
Dimensions (W*D*H)	6,058mm*2,438mm*2,896mm		6,058mm*2,438mm*2,896mm	
Weight	<16t		<25t	
Protection Degree of MV & LV Room	IP 54			
Operating Temperature Range	-30°C~40°C			
Relative Humidity	0~95% (Non-condensing)			
Operating Altitude	1,000m			
MV-LV AC Connections	Prewired and Pretested, No Internal Cabling Onsite			
LV & MV Room Cooling	Forced air cooling			
Communication protocols	Modbus TCP/IP			

□ Next-Gen Control



GridPoint Controller (GPC)

an advanced one-stop BESS control and status information and management center.

Power interface

AC400V/DC24V

Communication

Modbus TCP

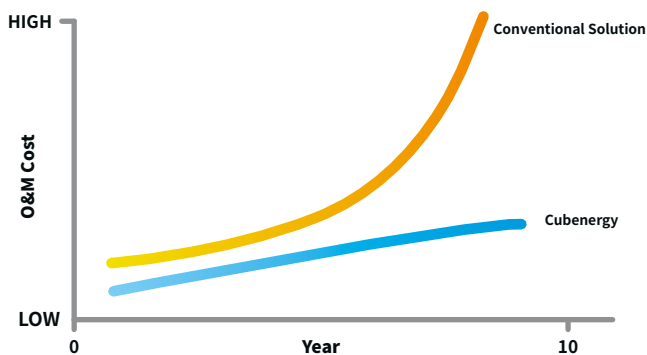
SPAC BMS

SPAC BMS delivers intelligent battery management through real-time cell voltage balancing. This proprietary technology minimizes energy loss caused by cell imbalances, increasing balance efficiency and extending battery cycle life. By ensuring uniform charge/discharge distribution across all cells, SPAC BMS maximizes usable capacity while maintaining stringent safety protocols—critical for large-scale BESS applications requiring long-term reliability.

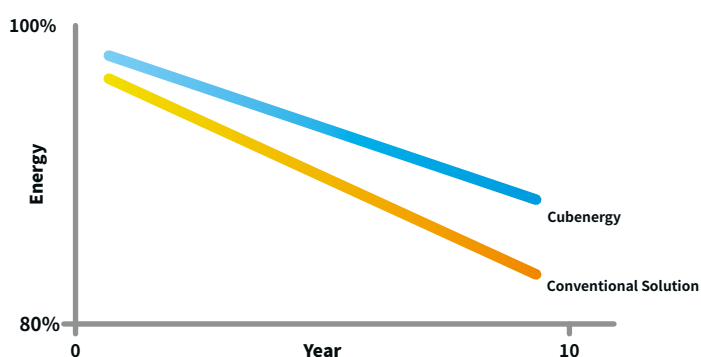
Functions

1. Monitor and protect safety of battery cell
2. Monitor and protect safety of electric system
3. Calculate and manage charging and discharging status
4. Calibrate and manage available energy
5. Optimize balance and manage cell consistency

Advantages



Maintenance Cost



Performance

□ Grid Forming*

UNICOM’ s BESS is equipped with advanced technologies for grid voltage support and frequency regulation. It employs Virtual Synchronous Generator (VSG) control to emulate the external characteristics of synchronous generators. This delivers essential inertia, damping, and dynamic grid support, ensuring robust and stable power system operation.



Superior Grid Compliance



Maximized ROI



Enhanced Grid Resilience

Functions

- **Inertia and Damping Control**
Inertia Emulation: The inverter simulates inertia using a virtual inertia time constant, allowing it to release or absorb energy during grid frequency or voltage changes, helping to slow down system fluctuations.
Damping Control: A virtual damping factor is applied to suppress oscillations and improve the dynamic stability of the grid.
- **Autonomous Frequency/Voltage Regulation**
Frequency Control: Dynamically adjusts active power based on frequency deviation (parameter kf: 0-200).
Voltage Control: Dynamically regulates reactive power based on voltage deviation (parameter kv: 0-100).
- **Multi-Scenario Adaptability**
Islanded Operation: Maintains voltage/frequency independently during grid outages for critical load supply.
Weak Grid Support: Stabilizes voltage/frequency in high-renewable penetration scenarios.
Renewables Coordination: Optimizes operation with wind/PV through power sharing.

Advantages



- **Enhanced Grid Stability**
Compensates for the lack of inertia in renewable energy sources (PV/wind), mitigates intermittent power fluctuations from PV/wind generation, improves grid absorption capacity and suppresses frequency deviations.



- **Multi-unit Cooperative Control**
Enables precise power allocation among parallel inverters via virtual impedance and adaptive algorithms, solving uneven power distribution issues in traditional droop control.



- **Flexible Operation Capability**
Supports both grid-connected and islanded modes, maintaining voltage/frequency in microgrids, and participates in grid frequency/voltage regulation in grid-tied application.



- **Power Quality Optimization**
Reduces current THD (Total Harmonic Distortion) to solve voltage imbalance issues caused by PV inverter grid integration.

*Seamless PQ Mode (Fixed Power Output) and GFM Mode (VSG Mode) Switching for Multi-Scenario Operation.